

Multiple Choice

1. **Answer:** D

Options: A) Error; B) a; C) b; D) c

Note: Correct option: D - c

2. **Answer:** C

Options: A) 4; B) 1; C) 10; D) 8

Note: Correct option: C - 10

3. **Answer:** D

Options: A) The wait is expected to be short.; B) A busy-wait loop is easier to code than an interrupt handler.; C) There is no other work for the processor to do.; D) The program executes on a time-sharing system.

Note: Correct option: D - The program executes on a time-sharing system.

4. **Answer:** D

Options: A) All bit strings whose number of zeros is a multiple of five; B) All bit strings starting with a zero and ending with a one; C) All bit strings with an even number of zeros; D) All bit strings with more ones than zeros

Note: Correct option: D - All bit strings with more ones than zeros

5. **Answer:** A

Options: A) 4; B) 1; C) 256; D) 16

Note: Correct option: A - 4

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' n^m '.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Round-robin'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'seed([x])'.

Long Form Response

11. **Answer:** `def get_max_occurring_char(str 1): | return ch`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def mul_list(nums 1,nums 2): | return list(result)`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key